

AQA A-Level Biology — Required Practicals

There are **12 required practicals** across the AQA A-Level Biology course (RP1–RP6 are also the AS required practicals). You will be tested on these in exams through questions about methods, apparatus, variables, analysis, evaluation, and the underlying biology. You do not need to have personally performed every practical, but you must understand them thoroughly.

Key practical skills tested: Independent, dependent and control variables; reliability (repeats, means); validity (controlling variables); accuracy and precision of measurements; sources of error and improvements; selecting appropriate apparatus; presenting data (tables, graphs); statistical analysis (chi-squared, t-test, Spearman rank correlation, standard deviation); drawing conclusions from data.

Table of Contents

1. [RP1: Effect of a Variable on Enzyme-Controlled Reaction Rate](#)
2. [RP2: Stained Squashes of Root Tip Cells \(Mitosis\)](#)
3. [RP3: Dilution Series & Calibration Curve for the Water Potential of Plant Tissue](#)
4. [RP4: Effect of a Variable on Membrane Permeability \(Beetroot\)](#)
5. [RP5: Dissection of Gas Exchange / Mass Transport System](#)
6. [RP6: Aseptic Technique & Antimicrobial Substances](#)
7. [RP7: Chromatography of Leaf Pigments](#)
8. [RP8: Effect of a Factor on Dehydrogenase Activity in Chloroplast Extracts](#)
9. [RP9: Effect of a Variable on the Respiration Rate of Single-Celled Organisms](#)
10. [RP10: Choice Chamber or Maze \(Animal Responses\)](#)
11. [RP11: Glucose Dilution Series & Colorimetric Calibration Curve](#)
12. [RP12: Distribution of a Species \(Quadrats & Transects\)](#)

RP1: Effect of a Variable on Enzyme-Controlled Reaction Rate

Aim: Investigate the effect of temperature (or pH, or substrate/enzyme concentration) on the rate of an enzyme-controlled reaction.

Biology link: Enzymes are biological catalysts (Topic 1). Their rate of reaction is affected by temperature (kinetic energy and denaturation), pH (ionisation of R groups in active site), and substrate concentration (collision frequency and saturation).

Apparatus

- Test tubes and rack, boiling tubes
- Catalase source (potato extract or liver homogenate) and hydrogen peroxide (H_2O_2) solution (typically 3-10 volume)
- Gas syringe (most accurate) or inverted measuring cylinder over water
- Delivery tube with bung (tight-fitting to prevent gas escape)
- Water baths at range of temperatures (e.g. 20, 30, 40, 50, 60°C) or buffer solutions for pH investigation
- Thermometer, stopwatch, graduated syringes or pipettes for measuring volumes

Method

1. Set up water bath at the required temperature. Place both enzyme and substrate solutions in separate test tubes in the water bath for **5 minutes** to equilibrate to the target temperature (this is a key step often missed).
2. Measure a fixed volume of enzyme and a fixed volume of H_2O_2 using graduated syringes.
3. Add the enzyme to the H_2O_2 , immediately fit the bung and delivery tube connected to the gas syringe (or inverted measuring cylinder).
4. Record the volume of O_2 gas collected every 30 seconds for 5 minutes.
5. Repeat at each temperature (or pH/concentration). Perform **at least 3 repeats** per condition and calculate mean values.
6. Calculate the **initial rate of reaction** from the steepest (linear) part of the volume-time curve (typically the first 30-60 seconds).

Alternative method (for amylase): Amylase breaks down starch. Mix amylase with starch solution at different temperatures. At regular intervals, take a sample and add to iodine solution. Time how long until iodine no longer turns blue-black (starch fully digested). Rate = $1/\text{time}$.

Variables

Independent	Dependent	Control Variables
Temperature (or pH, or concentration)	Volume of O ₂ produced per unit time (rate of reaction)	Volume and concentration of enzyme, volume and concentration of substrate, same enzyme batch, same total volume, equilibration time

Safety

- H₂O₂ is an irritant at higher concentrations — wear eye protection and gloves.
- Care with hot water baths — do not exceed 60°C; handle tubes with tongs.
- O₂ is released — ensure good ventilation. Do not approach with naked flames.
- If using buffer solutions, some are irritant — check MSDS.

Analysis

- Plot volume of O₂ (y-axis) against time (x-axis) for each condition — gives a product-time curve.
- Calculate **initial rate** = gradient of the tangent at time = 0 (or from the linear portion). Rate = $\Delta \text{volume} / \Delta \text{time}$ (cm³ s⁻¹ or cm³ min⁻¹).
- Plot initial rate (y-axis) against temperature or pH (x-axis) to find the **optimum**.
- For temperature: rate increases with temperature (more kinetic energy, more enzyme-substrate collisions) up to an optimum, then decreases sharply as the enzyme denatures (tertiary structure disrupted, active site shape changed permanently).
- For pH: bell-shaped curve with a peak at the optimum pH. Extreme pH values disrupt ionic bonds and hydrogen bonds in the enzyme, altering the active site shape.

Sources of Error and Improvements

Error	Effect	Improvement
Delay fitting bung after mixing	Gas escapes before collection begins; rate appears lower	Practise technique; use a side-arm flask with a rubber bung already in place
Temperature fluctuation during reaction	Rate varies inconsistently; enzyme may partially denature	Use a thermostatically controlled water bath; check temperature with thermometer
Enzyme concentration varies between batches	Unreliable comparisons between conditions	Use the same batch of enzyme preparation throughout; make one large batch and divide
Bubbles in delivery tube or syringe	Volume readings are inaccurate (artificially inflated)	Ensure delivery tube is filled with water before starting; check for leaks
Not equilibrating enzyme and substrate at target temperature	Reaction starts at wrong temperature; results invalid	Allow 5 minutes for both solutions to reach target temperature before mixing
Substrate depletion at end of reaction	Rate slows as substrate runs out, not due to variable being tested	Use initial rate (first 30–60 s) when substrate is in excess

Exam Tip: Always state that you would calculate the *initial rate* of reaction (not just "rate"). The rate at time = 0 avoids issues with substrate depletion, product inhibition, and enzyme inactivation. To calculate it: draw a tangent to the curve at $t = 0$ (or use the first linear portion) and measure its gradient. Also: when discussing temperature, distinguish between increased kinetic energy (reversible) and denaturation (irreversible — bonds broken, active site permanently changed).

RP2: Stained Squashes of Root Tip Cells (Mitosis)

Aim: Observe mitosis in root tip cells and calculate the mitotic index.

Biology link: The root tip meristem contains actively dividing cells (Topic 2). The mitotic index can indicate how rapidly a tissue is growing and has clinical relevance — a high mitotic index in tumour tissue indicates aggressive cancer.

Apparatus

- Garlic or onion root tips (grown in water for 3–5 days; garlic is often preferred as it has large chromosomes and a fast growth rate)
- Acetic orcein stain (or toluidine blue / Feulgen stain) — stains chromosomes dark red/purple
- 1 M hydrochloric acid, watch glass, water bath (or Bunsen burner with care)
- Microscope slides, coverslips, filter paper
- Optical microscope (x100 and x400 magnification), mounted needle, pencil eraser or thumb

Method

1. Cut approximately 2 mm of the **growing root tip** (the meristematic region where cells are actively dividing) and place on a watch glass.
2. Add a few drops of 1 M HCl and warm gently at ~60°C for 5 minutes in a water bath. This **hydrolyses the middle lamella** (breaks down pectin between cell walls), softening the tissue and separating cells so they form a single layer when squashed.
3. Carefully blot dry with filter paper, then add a few drops of **acetic orcein stain**. Leave for **10–15 minutes**. The stain binds to DNA, making chromosomes visible (darkly stained).
4. Transfer to a clean microscope slide. Add a coverslip. Place filter paper over the coverslip and **squash firmly** with a pencil eraser (press vertically, do not twist — this would smear cells). This spreads cells into a single layer.
5. View under microscope: start at **low power (x100)** to locate a good region of well-spread cells, then switch to **high power (x400)** to identify individual cells.
6. Identify and count cells in each stage of mitosis (prophase, metaphase, anaphase, telophase) and those in interphase. Count at least **200 cells** from multiple fields of view for reliability.

Safety

- HCl is corrosive — wear gloves and eye protection. Handle hot HCl with extra care.
- Acetic orcein stains skin and clothing permanently — handle carefully, wear gloves and lab coat.
- Care with hot water or Bunsen burner. Water bath is safer than direct heating.
- Sharp instruments (scalpels, needles) — cut away from body on a tile.

Analysis

Mitotic index = number of cells undergoing mitosis / total number of cells observed

- Record the number of cells at each stage: interphase, prophase, metaphase, anaphase, telophase.
- The **proportion of cells at each stage reflects the relative duration of that phase**.
Prophase is the longest phase of mitosis, so most dividing cells will be in prophase. Anaphase is the shortest, so fewest cells will be seen in this stage.
- A higher mitotic index indicates faster cell division (more cells in mitosis at any one time).
- Clinically, mitotic index is used to assess **tumour aggressiveness** — higher values indicate faster growth.

Sources of Error and Improvements

Error	Effect	Improvement
Cells overlapping after squash	Difficult to identify individual cells and stages	Squash more firmly (but do not twist); use thinner tip sections; ensure adequate HCl treatment
Insufficient staining time	Chromosomes not visible; difficult to identify stages	Leave stain for at least 10–15 minutes; ensure fresh stain is used
Counting non-meristematic cells	Lower mitotic index (mature cells do not divide)	Only cut and use the very tip (2 mm) where the meristem is located

Small sample size	Unreliable and unrepresentative mitotic index	Count at least 200 cells from multiple fields of view across the slide
Difficulty distinguishing phases	Misidentification of stages	Use reference images; practise identification; use higher magnification
Air bubbles under coverslip	Distort image; obscure cells	Lower coverslip at an angle; add more stain if needed

Exam Tip: Prophase typically has the most cells (longest phase), anaphase the fewest (shortest). Always state the formula: $\text{mitotic index} = \frac{\text{cells in mitosis}}{\text{total cells observed}}$. If asked to identify stages from a photograph: prophase = chromosomes visible but scattered; metaphase = chromosomes aligned at equator; anaphase = chromosomes being pulled apart to poles (V-shaped); telophase = two groups of chromosomes at poles, cell beginning to divide. Interphase cells show a dark nucleus with no visible individual chromosomes.

RP3: Dilution Series & Calibration Curve for the Water Potential of Plant Tissue

Aim: Make a dilution series of a solute (usually sucrose), use it to produce a calibration curve of percentage change in mass against concentration, and use the curve to find the water potential of a plant tissue such as potato.

Biology link: Osmosis and water potential (Topic 2). Water moves by osmosis from a region of higher (less negative) water potential to a region of lower (more negative) water potential across a partially permeable membrane. Pure water has a water potential of 0 kPa; adding solute makes it more negative. Where the tissue gains no mass and loses no mass, the tissue and the solution have the **same water potential**.

Apparatus

- Stock sucrose solution (e.g. 1.0 mol dm^{-3}) and distilled water
- Graduated pipettes or syringes, labelled boiling tubes and rack
- Potato (or other plant tissue), cork borer, scalpel, white tile, ruler
- Top-pan balance (reading to 0.01 g), paper towels for blotting, stopwatch
- A reference table or graph giving the water potential of sucrose solutions of known concentration

Making the Dilution Series

Use $C_1V_1 = C_2V_2$ to decide how much stock and water to mix. For 20 cm^3 of each solution from a 1.0 mol dm^{-3} stock:

Concentration (mol dm^{-3})	Stock sucrose (cm^3)	Distilled water (cm^3)
0.0	0	20
0.2	4	16
0.4	8	12
0.6	12	8
0.8	16	4
1.0	20	0

Method

1. Prepare the dilution series in labelled boiling tubes, with the **same volume** of solution in each tube. Mix each one thoroughly.
2. Use the cork borer to cut potato cylinders of the **same diameter**, then trim them with the scalpel to the **same length** (e.g. 3 cm). Remove any peel.
3. Blot each cylinder gently with a paper towel to remove surface water, then record its **initial mass**.

- Place one cylinder in each tube and start the stopwatch. Leave them all for the **same time** (e.g. at least 30 minutes, longer if possible) at the same temperature.
- Remove each cylinder, blot it in exactly the same way, and record its **final mass**.
- Carry out at least **3 repeats** at each concentration and calculate a mean percentage change in mass.

Variables

Independent	Dependent	Control Variables
Concentration of sucrose solution	Percentage change in mass of the tissue	Source of tissue (same potato), size of cylinders, volume of solution, time immersed, temperature, blotting technique

Safety

- Scalpels and cork borers are sharp — cut on a white tile, away from your body.
- Wipe up spilled sucrose solution straight away — it makes the floor and bench slippery and sticky.

Analysis

$$\text{Percentage change in mass} = (\text{final mass} - \text{initial mass}) / \text{initial mass} \times 100$$

- Use **percentage** change, not the change in grams, because the cylinders do not all start with exactly the same mass.
- Plot mean percentage change in mass (y-axis, with positive and negative values) against sucrose concentration (x-axis) and draw a line or curve of best fit. This is the **calibration curve**.
- Read off where the line crosses the x-axis (0% change). At this concentration there is no net movement of water, so the solution has the same water potential as the tissue.
- Convert that concentration into a water potential (kPa) using the reference table or graph for sucrose solutions.

- Below this concentration the tissue gains mass (the solution has a higher water potential than the cells); above it the tissue loses mass (the solution has a lower water potential).

Sources of Error and Improvements

Error	Effect	Improvement
Blotting inconsistently	Different amounts of surface water left on cylinders, so masses are inaccurate	Blot each cylinder the same way (e.g. roll once on a paper towel)
Cylinders of different size or from different potatoes	Different surface areas and starting water potentials	Use one cork borer and one potato; trim to the same length
Too short an immersion time	Tissue has not approached equilibrium, so changes are small and the intercept is less reliable	Leave cylinders for longer, keeping the time the same for every tube
Evaporation from uncovered tubes	Solutions become more concentrated	Cover the tubes (e.g. with cling film)
Wide gaps between concentrations	The x-intercept is found less precisely	Repeat with a narrower range of concentrations either side of the intercept

Exam Tip: Be ready to explain the calibration curve: the point of **no change in mass** is where the water potential of the solution equals the water potential of the tissue. Explain mass changes in terms of water potential (not "concentration of water"): in a dilute solution the solution has a **higher** water potential than the cells, so water enters the cells by osmosis. You may also be asked to calculate the volumes needed for a dilution using $C_1V_1 = C_2V_2$, and to explain why percentage change in mass is used.

RP4: Effect of a Variable on Membrane Permeability (Beetroot)

Aim: Investigate the effect of temperature (or alcohol concentration, or pH) on cell membrane permeability, using beetroot as a model system.

Biology link: Cell membrane structure (Topic 2). Beetroot cells contain the red-purple pigment **anthocyanin** in their vacuoles. The tonoplast (vacuolar membrane) and cell surface membrane normally retain this pigment inside the cell. When membranes are disrupted, anthocyanin leaks out into the surrounding solution. The amount of leakage (measured by colorimetry) indicates the degree of membrane damage.

Apparatus

- Fresh beetroot, cork borer (standardised size), white tile, sharp knife/scalpel
- Test tubes and rack, distilled water (measured volumes, e.g. 5 cm³)
- Water baths at range of temperatures (e.g. 20, 30, 40, 50, 60, 70°C)
- Colorimeter and cuvettes (using green or blue filter — complementary to red pigment)
- Thermometer, stopwatch, paper towels

Method

1. Use a cork borer to cut equal-sized beetroot cylinders (or use a ruler and knife to cut uniform discs of equal diameter and thickness). This standardises the **surface area** exposed.
2. **Rinse thoroughly** in cold distilled water until the water runs clear. This removes any anthocyanin released during cutting from damaged surface cells. This is critical — without adequate rinsing, all samples will appear to have high leakage.
3. Place one disc in each labelled test tube containing **5 cm³ distilled water** (measured precisely).
4. Place each tube in the appropriate water bath for a **standardised time** (e.g. exactly 20 minutes).
5. After exactly 20 minutes, remove the beetroot discs. Gently agitate each solution to distribute the pigment evenly.

6. Measure the **absorbance** of each solution using the colorimeter with a green/blue filter (zeroed with distilled water blank).
7. Repeat **3 times at each temperature** and calculate the mean absorbance.

Safety

- Sharp instruments (scalpel, cork borer) — cut on a white tile, away from the body.
- Hot water baths — handle tubes with tongs; take care with boiling water.
- Beetroot stains clothing and skin — wear lab coat and gloves if concerned.

Expected Results and Explanation

- At **low temperatures (20-30°C)**: low absorbance — membranes are intact, little pigment leaks out.
- At **moderate temperatures (30-40°C)**: slight increase — phospholipid bilayer becomes slightly more fluid, minor increase in permeability.
- At **high temperatures (above ~40°C)**: sharp increase in absorbance — membrane proteins **denature** (tertiary structure disrupted, proteins unfold and leave gaps in the membrane) AND phospholipids become much more fluid, increasing membrane permeability dramatically. Anthocyanin leaks freely.
- At very high temperatures (60-70°C): maximum leakage — membranes essentially destroyed.

Sources of Error and Improvements

Error	Effect	Improvement
Unequal disc sizes	Variable surface area affects amount of pigment released	Use same cork borer; cut all discs to uniform thickness with a ruler
Insufficient rinsing of cut surfaces	Surface pigment from damaged cells inflates all readings	Rinse until water runs completely clear before starting
Different time in water bath	Longer exposure = more leakage regardless of temperature	Use a stopwatch; stagger start times at 1-minute intervals; time precisely to 20 minutes each

Cells damaged during cutting	Extra pigment released from damaged cells (not due to temperature)	Discard end pieces; use only middle sections; handle gently; use a sharp borer
Pigment settling in tube	Uneven colour distribution; inconsistent readings	Agitate solution gently before transferring to cuvette

Exam Tip: Link results directly to membrane structure. At high temperatures: (1) **phospholipids gain kinetic energy** and move more, increasing membrane fluidity and creating gaps; (2) **membrane proteins denature** — hydrogen bonds and ionic bonds within the protein's tertiary structure are broken, the protein unfolds and can no longer maintain the membrane's barrier function. Mention both protein denaturation AND phospholipid fluidity for full marks. If asked about alcohol: ethanol dissolves phospholipids (it is a solvent for lipids), disrupting the bilayer structure and increasing permeability.

RP5: Dissection of Gas Exchange / Mass Transport System

Aim: Dissect an organism or organ to examine adaptations for gas exchange or mass transport (e.g. fish head/gills, heart, lungs, leaf).

Biology link: Gas exchange surfaces (Topic 3). Organisms have evolved specialised surfaces with features that maximise the rate of diffusion: large surface area, thin barrier (short diffusion distance), maintained concentration gradient (ventilation, blood supply).

Apparatus

- Fresh fish head (from fishmonger — not preserved), OR heart (lamb/pig), OR lungs
- Dissecting board, pins, scissors, forceps, scalpel, mounted needle
- Magnifying glass or dissecting microscope
- Paper towels, gloves, eye protection, lab coat

Method (Fish Gills)

1. Place the fish head on the dissecting board. Lift the **operculum** (gill cover) to expose the gills beneath.
2. Use scissors to carefully cut away the operculum, exposing the gill arches.
3. Identify the **4 gill arches** on each side, the **gill filaments** (primary lamellae — feathery pink structures), and the **gill rakers** (bony projections that filter food particles from water).
4. Remove one gill arch and place it in a dish of water on a white tile. Use a magnifying glass or dissecting microscope to observe the **secondary lamellae** (thin plate-like projections from each filament — the actual gas exchange surfaces).
5. Produce a **labelled biological drawing**: use clear, continuous lines; no shading; labels with ruled lines; include a title, magnification, and scale.

Safety

- Wear gloves when handling biological specimens (bacterial contamination risk).
- Sharp instruments — always cut away from the body on a dissecting board.
- Wash hands thoroughly with antibacterial soap after the practical.
- Dispose of biological material in designated biological waste bins (not regular bins).

Analysis — Structure-Function Relationships

Adaptation	How It Maximises Gas Exchange
Many gill filaments and lamellae	Provides a very large surface area for diffusion of O ₂ and CO ₂
Lamellae are very thin (one cell thick epithelium)	Short diffusion distance — gases cross quickly by diffusion
Dense network of capillaries in each lamella	Rich blood supply maintains a steep concentration gradient ; carries O ₂ away and brings CO ₂
Countercurrent flow (water and blood flow in opposite directions)	Maintains a concentration gradient along the entire length of the lamella — achieving ~80% O ₂ extraction (much more efficient than parallel flow)

Ventilation mechanism (buccal pumping)

Continuous flow of fresh water over gills maintains the concentration gradient

Exam Tip: In dissection questions, always link **structure to function**. For any gas exchange surface, mention: (1) large surface area, (2) thin barrier/short diffusion distance, (3) mechanism to maintain concentration gradient (ventilation + blood supply). For fish specifically, explain **countercurrent flow**: because blood flows in the opposite direction to water, there is always a diffusion gradient at every point along the lamella — this is far more efficient than if they flowed in the same direction (parallel flow), where equilibrium would be reached partway along.

RP6: Aseptic Technique & Antimicrobial Substances

Aim: Use aseptic technique to investigate the effect of antimicrobial substances (antibiotics or plant extracts) on bacterial growth.

Biology link: Infectious disease and antibiotics (Topic 3/6). Antibiotic resistance is a major global health concern. This practical demonstrates how antibiotics inhibit bacterial growth and how their effectiveness can be compared quantitatively.

Apparatus

- Nutrient agar plates (pre-poured and sterilised by autoclaving)
- Broth culture of bacteria (e.g. non-pathogenic *Escherichia coli* or *Micrococcus luteus*)
- Antibiotic discs (multidiscs or individual, e.g. penicillin, streptomycin, tetracycline) and/or plant extract soaked filter paper discs (e.g. garlic, mint, tea tree oil)
- Sterile inoculating loop or sterile spreader, Bunsen burner
- Forceps (to be flamed), marker pen (for labelling base of plate), adhesive tape, ruler, calculator

Aseptic Technique — Key Steps

- Work near a **lit Bunsen burner** — the updraft creates a convection current that carries airborne microorganisms upward, away from the work area (creating an **aseptic zone**).
- **Flame the inoculating loop** until red hot before and after use — this sterilises it by killing all microorganisms.
- **Flame the neck of culture bottles** briefly after opening and before closing — kills microorganisms at the rim.
- **Do not breathe on** or cough over open plates or cultures.
- Keep lids on plates as much as possible; only open when necessary.
- Wipe down the bench with disinfectant before and after the practical.

Method

1. Flame the inoculating loop, allow to cool briefly, dip into bacterial broth culture, and **spread evenly** across the agar plate surface using a zigzag pattern (streak plate method) or use a sterile glass spreader.
2. Using **flamed forceps**, place antibiotic discs (or plant extract discs) at equal distances on the agar surface. Press gently to ensure contact with the agar.
3. **Seal the plate** with two strips of adhesive tape in a cross pattern (do NOT seal completely around the edge — this could create **anaerobic conditions** inside the plate, promoting the growth of dangerous anaerobic pathogens).
4. Label the **base** of the plate (not the lid — lids can be accidentally swapped) with name, date, and disc identities.
5. Incubate the plate **upside down** (prevents condensation dripping onto colonies, which could spread bacteria and blur zones) at a **maximum of 25°C** for 24-48 hours.
6. After incubation, measure the **diameter of each clear zone of inhibition** (the area around each disc where bacteria have not grown) using a ruler. Do **NOT open the plate**.

Safety

- Incubate at **≤25°C in school laboratories** (not at 37°C — human body temperature would encourage growth of human pathogens that may have contaminated the plate).
- **Do not open plates after incubation** — unknown organisms may have grown.
- **Autoclave all plates** (121°C, 15 min) before disposal to kill all microorganisms.

- Maintain strict aseptic technique throughout to prevent contamination of the plate AND to prevent release of microorganisms into the laboratory.
- Wash hands with antibacterial soap before and after the practical.

Analysis

- Measure the **diameter** of each zone of inhibition (in mm). Take two measurements at right angles and average.
- Calculate the **area** of each zone: $A = \pi r^2$ (where $r = \text{diameter} / 2$). **Area gives a better comparison than diameter** — it measures the whole region where growth was inhibited, and it increases with the square of the radius, so differences between discs show up more clearly.
- Larger zone of inhibition = more effective antimicrobial substance (inhibits bacterial growth over a wider area).
- Subtract the area of the disc itself from the total zone area for a more accurate comparison.

Sources of Error and Improvements

Error	Effect	Improvement
Contamination from air	Unwanted colonies of environmental microorganisms grow on the plate	Work near a Bunsen burner; flame all equipment; minimise time plate lid is off
Uneven bacterial spread	Zones of inhibition are irregular or misleading	Use a sterile spreader with a turntable; rotate plate 90° between streaks
Discs placed too close together	Overlapping zones; cannot measure accurately	Space discs evenly (at least 2 cm apart); use a template or multipoint inoculator
Comparing diameters instead of areas	Underestimates differences in effectiveness (area scales with r^2)	Always calculate πr^2 for valid quantitative comparison

Inconsistent disc contact with agar

Poor diffusion; smaller zones than expected

Gently press each disc onto agar surface with sterile forceps to ensure good contact

Exam Tip: Two key points examiners look for: (1) Be able to calculate the **area** of a zone of inhibition from its diameter (πr^2 , remembering $r = \text{diameter} / 2$) — this calculation is commonly examined. (2) Know **why** plates are incubated at $\leq 25^\circ\text{C}$ in schools — to reduce the risk of growing human pathogens (which thrive at 37°C). In industrial/hospital labs, plates ARE incubated at 37°C because facilities are appropriate for handling pathogens safely.

RP7: Chromatography of Leaf Pigments

Aim: Separate and identify photosynthetic pigments from leaf extracts using thin-layer or paper chromatography, and compare the pigments in leaves of different plants (e.g. shade-tolerant and shade-intolerant plants, or leaves of different colours).

Biology link: Photosynthesis (Topic 5). Chloroplasts contain multiple pigments that absorb different wavelengths of light — this broadens the range of light energy that can be captured. Separating them demonstrates that leaves contain a mixture of pigments, not just chlorophyll.

Apparatus

- Fresh green leaves (spinach, nettle, or grass), pestle and mortar, clean sand (abrasive), small volume of propanone (acetone)
- Chromatography paper (or TLC plate with silica gel), pencil (NOT pen), ruler
- Boiling tube or chromatography jar with tight-fitting lid, capillary tube or micropipette
- Running solvent: propanone (acetone) or petroleum ether : acetone (9:1) mix

Method

1. Grind leaves with a small amount of clean sand (to break open cells and chloroplasts) and a few cm³ of propanone in a mortar. The sand acts as an abrasive. Filter the deep green extract through muslin or filter paper.
2. Draw a **pencil line** (origin line) 2 cm from the bottom of the chromatography paper. Use pencil because ink would dissolve in the solvent and interfere with results.
3. Apply leaf extract to the origin line using a capillary tube — build up a **concentrated spot** by applying small amounts repeatedly, allowing to dry between applications (5-10 applications).
4. Place the paper in the jar with the solvent level **below the origin line** (so the spot is above the solvent — otherwise pigments would dissolve directly into the solvent pool rather than being carried up by capillary action). Seal the jar with a lid to prevent solvent evaporation.
5. Allow the solvent to rise by capillary action until it is approximately 1-2 cm from the top. **Mark the solvent front** immediately with pencil before it evaporates.
6. Remove the paper and allow it to dry. Measure the distance from the origin line to the centre of each pigment spot, and the distance from the origin to the solvent front.

Safety

- Propanone (acetone) is **highly flammable** and produces irritating fumes — NO naked flames near the experiment; work in a well-ventilated area or fume cupboard.
- Petroleum ether is extremely flammable — same precautions.
- Avoid prolonged skin contact with solvents — wear nitrile gloves.

Analysis

$$R_f = \text{distance moved by pigment from origin} / \text{distance moved by solvent front from origin}$$

- Calculate R_f value for each separated pigment. Compare to reference/published R_f values to **identify each pigment**.
- Typical order of separation (highest R_f = most soluble in the non-polar solvent, travels furthest):

Pigment	Colour	Approximate R_f	Solubility
---------	--------	-------------------	------------

Carotene	Orange-yellow	~0.95	Most soluble (non-polar, dissolves readily in non-polar solvent)
Xanthophyll	Yellow	~0.7	Moderately soluble
Chlorophyll a	Blue-green	~0.5	Less soluble
Chlorophyll b	Yellow-green	~0.4	Least soluble (more polar; held more strongly by the stationary phase)

Sources of Error and Improvements

Error	Effect	Improvement
Origin line below solvent level	Pigments dissolve directly into solvent and are washed away	Ensure the solvent level is 1 cm below the origin line
Jar lid not sealed properly	Solvent evaporates; uneven solvent front; inconsistent R_f values	Use a tight-fitting lid or cover with cling film; line jar with filter paper soaked in solvent to saturate the atmosphere
Extract not concentrated enough	Spots too faint to see or measure	Apply many small applications, drying between each; use a concentrated extract
Using pen instead of pencil for origin	Pen ink dissolves in solvent and runs up the paper, contaminating results	Always use a pencil line
Not marking solvent front immediately	Solvent evaporates; R_f values cannot be calculated accurately	Mark the solvent front with pencil the instant you remove the paper

Exam Tip: Carotene has the highest R_f because it is the most non-polar pigment and therefore most soluble in the non-polar organic solvent. Chlorophyll b has the lowest R_f because it is the most polar (it has more polar -CHO groups compared to chlorophyll a's -CH₃ group) and is more attracted to the polar stationary phase (paper/silica). R_f values are specific to a given solvent system and temperature — always state the solvent used. When comparing plants, shade-tolerant species often have relatively more chlorophyll b and other accessory pigments, which widen the range of wavelengths they can absorb in dim light.

RP8: Effect of a Factor on Dehydrogenase Activity in Chloroplast Extracts

Aim: Investigate the effect of a named factor (e.g. light intensity, temperature or a herbicide) on the rate of dehydrogenase activity in a suspension of isolated chloroplasts, using the redox indicator DCPIP.

Biology link: The light-dependent reaction (Topic 5). In the thylakoid membranes, electrons released from chlorophyll are passed along an electron transfer chain and are normally used, with protons, to reduce NADP. **DCPIP** is an artificial electron (hydrogen) acceptor that can take the place of NADP. It is **blue when oxidised** and becomes **colourless when reduced**, so the faster the blue colour disappears, the faster the dehydrogenase activity in the chloroplasts.

Apparatus

- Fresh green leaves (e.g. spinach), scissors, chilled pestle and mortar (or blender)
- Ice-cold **isolation medium** (a buffered solution with the same water potential as the chloroplasts, e.g. buffered sucrose solution)
- Muslin or nylon mesh, funnel, centrifuge tubes, centrifuge, beaker of ice
- DCPIP solution, test tubes, graduated pipettes or syringes
- Bench lamp, metre rule, aluminium foil, stopwatch
- Colorimeter with a red filter and cuvettes (optional, for quantitative results)

Method

1. Cut the leaves into small pieces (removing thick midribs) and grind them in a chilled mortar with a small volume of ice-cold isolation medium.
2. Filter the mixture through several layers of muslin into a chilled centrifuge tube kept in ice, to remove cell debris.
3. Centrifuge the filtrate so that the chloroplasts form a pellet. Pour off the supernatant and resuspend the pellet in a small volume of ice-cold isolation medium. Keep this **chloroplast suspension** in ice.
4. Set up test tubes each containing the same volume of chloroplast suspension and the same volume of DCPIP. Place them at different distances from the lamp (e.g. 10, 20, 30, 40 cm) to vary light intensity.
5. Set up **controls**: a tube of chloroplast suspension and DCPIP wrapped in foil (no light), and a tube of isolation medium and DCPIP with no chloroplasts, placed in the light.
6. Time how long each tube takes to lose its blue colour, or measure absorbance with a colorimeter at regular intervals. Zero the colorimeter with chloroplast suspension and isolation medium but no DCPIP, so that only the blue of the DCPIP is being measured.
7. Repeat at least 3 times at each distance and calculate means.

Why the Isolation Medium Is Ice-Cold, Buffered and Isotonic

Feature	Reason
Ice-cold	Slows the activity of enzymes that would damage or break down the chloroplasts
Buffered	Keeps the pH constant so chloroplast proteins, including enzymes, are not denatured
Same water potential as the chloroplasts	Prevents net movement of water by osmosis, so chloroplasts do not swell and burst or shrink

Variables

Independent	Dependent	Control Variables
Light intensity (distance from lamp), or another named factor such as temperature or herbicide concentration	Time for DCPIP to decolourise, or rate of change of absorbance	Volume and concentration of chloroplast suspension and DCPIP, temperature, same leaf batch, background light

Safety

- DCPIP can irritate skin and eyes — wear eye protection and avoid skin contact.
- Lamps get hot — do not touch the bulb, and keep water and solutions away from electrical equipment.
- Balance the centrifuge tubes and keep the lid closed until the rotor has stopped.
- Take care with scissors and glassware.

Analysis

- Rate of dehydrogenase activity = $1 / \text{time taken for DCPIP to decolourise (s}^{-1}\text{)}$, or the gradient of absorbance against time.
- Light intensity is inversely proportional to the square of the distance from the lamp (light intensity $\propto 1/d^2$), so you can plot rate against $1/d^2$.
- Expect the rate to increase with light intensity, because more electrons are excited and passed to DCPIP, until another factor becomes limiting.
- The foil-covered tube should stay blue, showing that light is needed. The tube without chloroplasts should stay blue, showing that the chloroplasts (not the light alone) reduce the DCPIP.

Sources of Error and Improvements

Error	Effect	Improvement
Judging the end point by eye	Subjective; different people choose different end points	Use a colorimeter, or compare with a colour standard

Heat from the lamp	Temperature varies with distance, a second variable affecting rate	Place a heat shield (e.g. a beaker of water) between lamp and tubes; use an LED lamp
Chloroplasts damaged during extraction	Low or variable activity	Keep everything ice-cold; use buffered, isotonic medium; work quickly
Background light in the room	Light intensity is not only controlled by the lamp	Carry out the investigation in a darkened room
Chloroplasts settling in the tubes	Uneven light exposure and colour	Gently mix each tube before taking readings

Exam Tip: Learn the colour change: DCPIP goes from **blue to colourless when it is reduced** (it accepts electrons that would otherwise reduce NADP). Be able to give the purpose of each control tube, and the three features of the isolation medium with a reason for each. If a herbicide blocks the electron transfer chain, fewer electrons reach DCPIP, so it stays blue for longer.

RP9: Effect of a Variable on the Respiration Rate of Single-Celled Organisms

Aim: Investigate the effect of a named variable (e.g. temperature or respiratory substrate) on the rate of respiration of a culture of single-celled organisms such as yeast, using a respirometer to measure O_2 consumption or a redox indicator.

Biology link: Respiration (Topic 5). All living organisms respire continuously. The rate of O_2 consumption is a direct measure of aerobic respiration rate. Temperature affects enzyme activity (respiratory enzymes in glycolysis, Krebs cycle, and ETC).

Apparatus

- Simple respirometer (two tubes connected to a capillary tube/manometer with a scale)
- Organism: a culture of a single-celled organism, e.g. yeast (*Saccharomyces cerevisiae*) suspended in glucose solution. (Respirometers are also used with germinating seeds or small invertebrates, but this AQA practical specifies single-celled organisms.)
- Soda lime or potassium hydroxide pellets (to absorb CO_2 produced by respiration)
- Capillary tube with graduated scale, coloured oil or fluid drop (indicator of gas volume change)
- Water bath (for temperature control), thermometer, stopwatch
- Glass beads or boiled (dead) yeast suspension for the control tube — same volume, no living cells

How the Respirometer Works

The key principle: soda lime (or KOH) **absorbs all CO_2** produced by the organisms. Therefore, the only gas change is the **consumption of O_2** . As O_2 is used up, the gas volume in the sealed tube decreases, and the **coloured fluid drop moves towards the organisms**. The distance moved per unit time = rate of O_2 uptake = rate of aerobic respiration.

The **control tube** (containing glass beads + soda lime, but no living organisms) accounts for any changes in gas volume due to temperature or atmospheric pressure changes that are NOT due to respiration.

Method

1. Set up the respirometer: place a fixed volume of yeast suspension in glucose solution in one tube, with soda lime held separately (e.g. in a gauze basket or behind cotton wool) so it cannot touch the culture. Set up a **control tube** with the same volume of boiled yeast suspension (or glass beads) and soda lime.
2. Seal both tubes and connect them to the manometer/capillary tube via airtight connections (use Vaseline on joints to prevent leaks).
3. Place in a water bath at the desired temperature. Allow **5 minutes to equilibrate** so the temperature stabilises and the yeast reaches a steady rate of respiration.
4. Introduce a coloured fluid drop (or oil droplet) into the capillary tube using a syringe.

5. Record the **distance the fluid drop moves** towards the organisms every minute for 10–15 minutes.
6. Repeat at different temperatures (e.g. 15, 20, 25, 30, 35, 40°C). Calculate rate of O₂ uptake (mm min⁻¹ or mm³ min⁻¹ if capillary tube cross-section is known).

Safety

- Soda lime and KOH are **corrosive** — wear gloves and eye protection. Avoid direct contact with organisms.
- Use aseptic technique when handling yeast cultures, and disinfect or autoclave cultures before disposal.
- Care with water baths and glassware; keep temperatures within a sensible range (e.g. 15–40°C).

Analysis

- **Rate of O₂ consumption** = distance moved by fluid drop / time (mm min⁻¹).
- Plot rate of O₂ uptake (y-axis) against temperature (x-axis). Expect a general increase with temperature (Q₁₀ effect on enzyme activity) up to an optimum, then a decrease at high temperatures (enzyme denaturation).
- The **control tube** corrects for any volume changes due to atmospheric pressure and temperature fluctuations. If the control drop moves, subtract this from the experimental reading.

Alternative method (redox indicator): Add a fixed volume of a redox indicator such as methylene blue to yeast in glucose solution in a water bath. Dehydrogenase enzymes in respiration remove hydrogen from substrates, and the indicator accepts it, turning from blue to colourless. Time how long the colour takes to disappear; rate = 1/time. A tube of boiled yeast acts as the control.

Sources of Error and Improvements

Error	Effect	Improvement
-------	--------	-------------

Leaks in apparatus	Air enters; fluid drop does not move correctly (or moves the wrong way)	Check all connections; apply Vaseline to all joints; test for leaks before starting
Soda lime exhausted or insufficient	CO ₂ not fully absorbed; gas volume change is less than expected (CO ₂ partially replaces O ₂)	Use fresh, excess soda lime for each experiment
Temperature fluctuation in water bath	Affects gas expansion/contraction independently of respiration	Use a thermostatically controlled water bath; allow adequate equilibration time
Yeast not yet respiring at a steady rate	Unrepresentative early readings	Allow a 5-minute equilibration period after setup before starting measurements
Yeast settling to the bottom of the tube	Cells in the sediment receive less oxygen and glucose, so the rate is inconsistent	Swirl or stir the suspension gently before and during readings; use the same volume and density of culture each time

Exam Tip: The soda lime absorbs CO₂, so **only O₂ consumption** causes the gas volume change. Without soda lime, the CO₂ produced would partially replace the O₂ consumed, and the volume change would be much smaller (depending on the RQ of the substrate). The control tube is essential — always explain its purpose: it accounts for any volume changes due to factors other than respiration (atmospheric pressure changes, temperature fluctuations). To measure anaerobic respiration, you would remove the soda lime and measure CO₂ production instead.

RP10: Choice Chamber or Maze (Animal Responses to Stimuli)

Aim: Investigate the effect of an environmental variable (light/dark, humidity) on the movement of an animal (e.g. woodlice or maggots) using either a choice chamber or a maze.

Biology link: Organisms respond to stimuli (Topic 6). Woodlice show **kinesis** (non-directional response — they change their speed or rate of turning in response to stimulus intensity) rather than taxis (directional movement towards or away from a stimulus). They tend to move slower and turn less in favourable conditions (dark, humid), accumulating there over time.

Apparatus

- Choice chamber (circular or rectangular, divided into 2-4 compartments with a perforated divider and clear lid)
- 10-20 woodlice (collected humanely from under logs, stones, or leaf litter)
- Damp cotton wool (for humid conditions) and dry silica gel / anhydrous calcium chloride (desiccant, for dry conditions)
- Black card or aluminium foil (for dark conditions), lamp (for light conditions)
- Stopwatch, data recording sheet

Method

1. Set up **two contrasting environments** in the choice chamber. Test **only one variable at a time**:
 - For humidity: one side has damp cotton wool (humid), the other has dry silica gel (dry) — both kept at the same light level
 - For light: one side is covered with black card (dark), the other is exposed to a lamp (light) — both kept at the same humidity
2. Place 10 woodlice in the **centre** of the chamber (on the divider), ensuring they have equal access to both sides.
3. Replace the lid and start the timer immediately.
4. Count the number of woodlice on each side **every 2 minutes for 20-30 minutes**. Record in a table.
5. Repeat the entire experiment **at least 3 times** with different woodlice. Calculate mean number on each side at each time interval.

Maze alternative: Place an animal at the start of a simple T-maze with contrasting conditions in each arm (e.g. dark and light). Record which arm it turns into, repeat with many individuals (or many trials), and analyse the turns with a chi-squared test.

Safety and Ethics

- Handle woodlice **gently** — they are living organisms deserving ethical treatment.
- **Return all woodlice** to their original habitat after the experiment.
- Do not expose to extreme conditions (very hot, direct strong light) for extended periods.
- Wash hands after handling. Silica gel can be irritating — avoid direct skin contact.

Analysis

- Calculate the **mean number** of woodlice on each side at each time interval across all repeats.
- Woodlice show **kinesis**: they move more slowly and turn less in preferred conditions (dark, humid), so they **accumulate** in the dark/humid side over time — not because they are attracted to it, but because they spend more time there once they arrive.
- Use a **chi-squared test** to determine if the final distribution is significantly different from the 50:50 distribution expected if woodlice showed no preference (null hypothesis: no significant difference from a 1:1 distribution).

Sources of Error and Improvements

Error	Effect	Improvement
Woodlice on the divider/boundary	Difficult to assign to one side	Decide a consistent rule beforehand (e.g. assign by head position); exclude borderline individuals from count
Two variables changed at once (e.g. dark AND humid on same side)	Cannot determine which factor causes the preference	Only vary ONE factor per experiment; keep all other conditions identical
Small sample size / few repeats	Results may not be statistically significant or	Use at least 10 woodlice per trial; repeat at least 3 times; pool data

	representative	from class
Temperature differences between sides	Confounding variable — woodlice may respond to temperature, not the intended variable	Ensure both sides are at the same temperature (no lamp heat on one side only — use a distant lamp or LED)
Organisms stressed by handling	Erratic initial behaviour unrelated to the variable	Allow a 2-minute settling period before starting to count

Exam Tip: Woodlice show **kinesis, not taxis**. Kinesis is a **non-directional** response where the organism changes its **speed of movement** or **rate of turning** depending on stimulus intensity. In unfavourable conditions (light, dry), they move faster and turn more frequently (increasing the chance of finding better conditions). In favourable conditions (dark, humid), they slow down and turn less — so they accumulate there. This is different from taxis, which is a directional movement towards or away from a stimulus (e.g. positive phototaxis in *Euglena*).

RP11: Glucose Dilution Series & Colorimetric Calibration Curve

Aim: Produce a dilution series of glucose solutions, use colorimetry to make a calibration curve, then use the curve to find the glucose concentration of an unknown 'urine' sample.

Biology link: Control of blood glucose and diabetes (Topic 6). In uncontrolled diabetes, blood glucose can rise so high that not all glucose is reabsorbed in the kidney, so glucose appears in urine. Colorimetry turns the colour of a Benedict's test into a number, so the glucose concentration can be measured rather than just estimated from the colour.

Apparatus

- Stock glucose solution of known concentration and distilled water
- Graduated pipettes or syringes (accurate to 0.1 cm^3), test tubes and rack
- Benedict's reagent and a water bath at about 80°C (or a boiling water bath)

- Filter paper and funnels, or a centrifuge, to remove the precipitate
- Colorimeter with a **red filter** and clean cuvettes
- Unknown 'urine' sample (a glucose solution of unknown concentration)

Method

1. **Prepare a dilution series** of glucose. Either make a serial dilution (e.g. transfer 5 cm^3 of stock to 5 cm^3 of water, mix, and repeat, halving the concentration each time) or mix stock and water in different proportions to give evenly spaced concentrations. Include a tube of distilled water only (0 concentration).
2. Add the **same volume of Benedict's reagent** to the same volume of each solution, including the unknown. Benedict's must be in excess so that the highest concentration does not use it all up.
3. Heat all the tubes together in the water bath for the **same time** (e.g. 5 minutes), then cool them.
4. Remove the brick-red precipitate by filtering or centrifuging, so that the **remaining blue solution** can be measured. The more glucose there was, the more copper(II) ions were reduced to precipitate, so the less blue the remaining solution.
5. **Zero the colorimeter** with distilled water, then measure the absorbance of each solution using a red filter (blue solutions absorb red light most strongly).
6. Plot **absorbance (y-axis) against glucose concentration (x-axis)** to produce a **calibration curve**.
7. Measure the absorbance of the treated unknown 'urine' sample and read its concentration off the calibration curve.

Safety

- Benedict's reagent contains copper sulfate — irritant. Wear eye protection.
- Hot water baths — use tongs or a test-tube holder; do not overfill tubes.
- Treat the 'urine' sample as a laboratory solution — never use real urine in a school laboratory.

Analysis

- With the precipitate removed, absorbance **falls** as glucose concentration rises, because less blue copper(II) sulfate is left in solution. Draw a smooth line or curve of best fit through the points.
- Read the unknown concentration: draw a **horizontal line from its absorbance value to the calibration curve**, then a **vertical line down to the x-axis**. Read the concentration.
- If the unknown's absorbance falls outside the calibration range, dilute it by a known factor, test it again, then multiply the result by the dilution factor.
- Only read values **within** the range of your known standards (interpolate); do not extend the curve beyond them.

Sources of Error and Improvements

Error	Effect	Improvement
Air bubbles in cuvette	Scatter light; give incorrect absorbance readings	Tap cuvette gently to dislodge bubbles; wipe with lens tissue
Inaccurate pipetting during dilution	Dilution concentrations wrong; calibration curve unreliable	Use calibrated graduated pipettes; read at eye level; practise technique
Fingerprints or scratches on cuvette	Light absorbed/scattered by contaminants	Handle cuvettes by the ridged sides or top edges only; always orient the cuvette the same way
Different heating times or temperatures	Different amounts of precipitate form, independent of glucose concentration	Heat all tubes together in the same water bath for the same time
Precipitate left in the cuvette	Particles scatter light and give unreliable readings	Filter or centrifuge before measuring; let any remaining solid settle
Not mixing dilutions thoroughly	Concentration gradient within tube; inaccurate readings	Invert tube 3-4 times after each dilution step

Exam Tip: A serial dilution gives a wide range of concentrations in equal ratio steps; each 1-in-2 dilution halves the concentration. Always keep the volume transferred and the volume of water the same at each step. Know which way your calibration curve slopes: if you measure the **blue solution left after removing the precipitate**, higher glucose gives **lower** absorbance (higher % transmission). The unknown must be treated in exactly the same way as the standards (same volumes, same heating time) for its reading to be valid.

RP12: Distribution of a Species (Quadrats & Transects)

Aim: Investigate the effect of a named environmental factor (e.g. light intensity or soil moisture) on the distribution of a given species, using appropriate sampling techniques.

Biology link: Ecology and biodiversity (Topics 4 and 7). The distribution of organisms is influenced by both abiotic factors (light, temperature, soil pH, moisture, exposure) and biotic factors (competition, predation). Sampling must be done appropriately to give a representative picture of the community.

Apparatus

- Quadrats (e.g. 0.5 m x 0.5 m frame, possibly subdivided into 25 smaller squares for % cover estimation)
- Tape measure (30 m or longer) for transect line
- Random number generator or random number table (for random quadrat placement)
- Species identification keys, recording sheets, pencil
- Abiotic measurement tools: light meter, soil pH probe/meter or BDH soil test kit, soil moisture meter, thermometer, anemometer (wind speed)

Two Sampling Approaches

Method	When to Use	How
--------	-------------	-----

Random quadrat sampling	Estimating the abundance/density of species in a uniform habitat (e.g. a field)	Use random number pairs as x,y coordinates to position each quadrat. This avoids sampling bias. Count organisms or estimate % cover in each quadrat. Repeat many times (at least 10–20).
Systematic sampling along a transect	Studying changes in species distribution along an environmental gradient (e.g. from seashore to cliff top, from woodland edge to interior, from path to field centre)	Lay a tape measure along the gradient. Place quadrats at regular intervals (e.g. every 2 metres). Record species and abundance + abiotic factors at each position.

Method (Interrupted Belt Transect Example)

1. Lay a tape measure in a straight line across the habitat, along the environmental gradient of interest (e.g. from the high water mark up a rocky shore).
2. Place a quadrat at regular intervals along the transect (e.g. every 2 metres, or every 5 metres for longer transects).
3. At each quadrat position:
 - Identify and record all species present (use identification keys)
 - Estimate % cover for each plant species (or count individuals for animals)
 - Measure relevant abiotic factors at the same point (light intensity, soil pH, soil moisture, temperature, wind speed)
4. Record all data in a prepared table. Repeat the transect in parallel (if possible) for reliability.

Safety

- Conduct a **risk assessment** for fieldwork: assess terrain (uneven ground, cliffs, slippery rocks), weather conditions, tidal risks (on a seashore), and lone working policy.
- Work in pairs or groups. Carry a mobile phone. Inform someone of your location and expected return time.
- Wash hands after handling soil (may contain tetanus bacteria, Weil's disease from rat urine).

- Be aware of hazardous plants (stinging nettles, thorns, giant hogweed) and animals (ticks, adders).
- Wear appropriate clothing and footwear for the habitat.

Analysis

- Calculate **species density** = mean count per quadrat / quadrat area (number per m²).
- Calculate **% frequency** = (number of quadrats containing the species / total number of quadrats) x 100.
- Use a **kite diagram** or bar chart to display species distribution along a transect.
- If several species are recorded, calculate the AQA **index of diversity**: $d = N(N - 1) / \sum n(n - 1)$, where N = total number of organisms of all species and n = number of organisms of each species.
- Use **Spearman's rank correlation** to test for a significant correlation between species abundance and an abiotic factor (e.g. "Is there a significant correlation between % cover of a plant species and light intensity?").
- Spearman's r_s ranges from -1 (perfect negative correlation) through 0 (no correlation) to +1 (perfect positive correlation). Compare r_s to the critical value for your sample size at $p = 0.05$.

Sources of Error and Improvements

Error	Effect	Improvement
Non-random placement of quadrats (e.g. choosing "interesting" spots)	Sampling bias — results not representative of the whole habitat	Use a random number table/generator for coordinates; place quadrats without looking
Too few quadrats	Unrepresentative data; poor statistical power	Use at least 10-20 quadrats per area; pool class data
Misidentification of species	Inaccurate species count and diversity calculation	Use reliable identification keys; take photographs for verification; have an expert check

Seasonal/time-of-day variation	Different results at different times	Sample at the same time of day and same season for valid comparisons
Trampling or disturbing the habitat	Alters the community being studied	Approach carefully; use a random walk pattern; minimise time in each area

Exam Tip: Random sampling avoids bias — always explain how you would achieve randomness (random number coordinates from a baseline and side-line). For **transects**, explain why systematic sampling is appropriate: you want to detect changes in distribution along a gradient, which random sampling might miss. Know when to use each statistical test: **chi-squared** for comparing observed vs expected frequencies (categories); **Spearman's rank correlation** for testing the relationship between two continuous variables (e.g. species abundance vs distance along transect); **Student's t-test** for comparing means of two groups.